

Learned Optimal Visual Time-of-Flight Imaging With Fisher Information Guidance

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Abstract—Indirect time-of-flight (iToF) imaging provides absolute depth by encoding light transport delays into correlation measurements via active illumination modulation and demodulation. This technology plays a crucial role in object detection and scene understanding and has been widely adopted in applications such as robotics, automotive driving, and augmented reality. However, iToF systems often suffer from low signal-to-noise ratio (SNR), particularly under strong ambient illumination. Signal attenuation and noise-induced phase errors severely degrade depth accuracy, while reliable depth discontinuity and edge reconstruction remain challenging. In this paper, we introduce an optimal visual ToF imaging scheme, optimized through end-to-end learning of iToF coding functions and depth reconstruction, guided by discriminative Fisher information supervision. By integrating vision information from RGB images, we facilitate the convergence of the optimization for iToF coding functions and enhance robustness under noisy inputs. Moreover, we design a dual-branch visual ToF reconstruction network that estimates depth from iToF measurements while exploiting edge guidance derived from visual information to preserve fine geometric details. Extensive experiments on both synthetic and real-world datasets verify the effectiveness of the proposed approach, particularly in challenging low-SNR conditions.

Index Terms—Visual ToF imaging, end-to-end optimization, discriminative Fisher information.

I. INTRODUCTION

TIME-OF-FLIGHT (ToF) sensing enables direct and efficient depth acquisition and has therefore been widely adopted in practical applications, including 3D reconstruction, human–computer interaction, and autonomous driving. ToF imaging technologies can be mainly classified into direct ToF

(dToF) and indirect ToF (iToF). dToF sensors measure depth by directly timing the round-trip travel of emitted light pulses, typically implemented using time-to-digital converters (TDCs). In contrast, iToF estimates depth by measuring the correlation between modulated illumination and on-sensor demodulation. Depth is encoded in samples of the correlation between the scene response function and a set of correlation functions, which are determined by the source modulation and sensor demodulation waveforms. The design of these functions is therefore key to solving the inverse problem of depth recovery [1]. Compared with dToF, this coded modulation–demodulation framework introduces a design space that can be optimized for specific imaging conditions by tailoring the demodulation waveform and coding strategy.

In the past decade, many commercial depth cameras have been developed, such as Kinect-v2 [2], PMD [3], and SoftKinect [4]. Despite substantial progress in resolution and accuracy in recent years, the depth images captured by these commercial depth sensors still suffer from severe noise. Therefore, developing a high-precision, noise-robust 3D sensing system is of great significance to both academia and industry.

In iToF imaging, strong ambient light often degrades the signal-to-noise ratio (SNR), resulting in substantial reconstruction artifacts. Adopting a higher energy light source, increasing the exposure time, or capturing more measurements could improve the SNR of detection, at the expense of increasing the power consumption or the acquisition time. Recently, Su et al. [5] proposed a ToF reconstruction network to improve robustness against noise through post-processing, without considering the influence of coding schemes. Traditional coding schemes, such as sinusoidal or square-wave modulation, while widely adopted, suffer from aliasing and perform poorly at low SNR. Gupta et al. [6] and Gutierrez-Barragan et al. [7] designed a series of Hamilton coding functions that outperform the commonly adopted sinusoidal and square coding functions. However, because of the assumption of small noise, their performance in low-SNR scenarios remains limited.

In our previous work [8], we introduced a Fisher-information-guided learned iToF imaging framework for coding-function optimization and depth estimation. From a compressed sensing perspective, by formulating the correlation functions in iToF as a sensing matrix, coherence has been proposed to characterize the recoverability of depth signals [9], [10] and to provide an upper bound on information loss [11]. To address practical low-SNR scenarios, we propose to optimize the iToF coding functions

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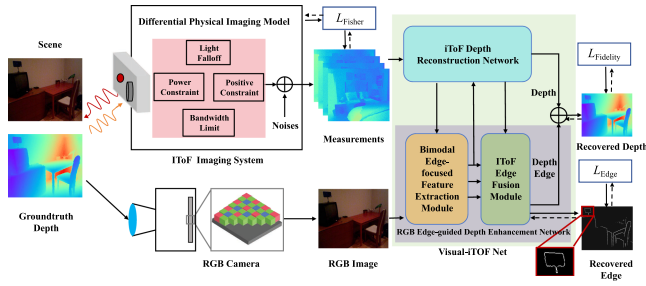


Fig. 1. Overview of the proposed learned optimal visual ToF imaging framework. The framework combines iToF imaging and RGB imaging, utilizing a dual-branch depth reconstruction network to recover depth from the captured measurements. End-to-end optimization of the visual ToF scheme ensures robust depth imaging performance, especially under large noise scenarios.

using Fisher information, which quantifies the information content of measurements under practical noise statistics and specific scene conditions. Based on this foundation, we present a novel visual ToF imaging framework that leverages visual information to enhance both the learning of iToF coding functions and the accuracy of depth reconstruction. The proposed visual ToF network fuses learned iToF cues with complementary visual guidance, thereby improving robustness and overall reconstruction quality.

The overall visual ToF imaging scheme is shown in Fig. 1. The visual ToF imaging scheme comprises both differential iToF imaging with learnable coding functions and RGB imaging. Based on iToF measurements and an RGB image, a dual-branch visual ToF depth reconstruction network is proposed, and the entire visual ToF imaging scheme is optimized end-to-end. The superiority of the proposed visual ToF imaging system is demonstrated through extensive experiments, including simulations and real-world applications, conducted using the constructed prototype system.

In particular, we make the following contributions:

- We *propose* a visual ToF imaging scheme to jointly optimize the coding functions of iToF and the depth reconstruction with visual information guidance under large noise scenarios.
- We *design* a dual-branch visual ToF depth reconstruction network that effectively extracts the information from the iToF measurement and visual information to realize high-accuracy depth reconstruction.
- We *introduce* the supervision mechanism with discriminative Fisher loss to maximize the encoded depth information of the proposed visual ToF imaging.
- We *build* a prototype visual ToF system with the learned optimal coding functions and *demonstrate* the state-of-the-art performance of the proposed visual ToF imaging method, both in simulation and in real data.

II. RELATED WORK

A. iToF Imaging

In high-SNR scenarios, conventional iToF cameras [1], [12] that use sinusoidal or square wave coding functions can achieve

high accuracy. However, as the SNR decreases, the reconstruction error increases with conventional iToF cameras. It is easier and more affordable to modify the coding functions compared to improving the performance of the sensor device. Many works have been offered to construct various coding functions for ToF imaging over the years. Payne et al. [13] proposed to reduce the duty cycle of square and sinusoid functions in order to suppress aliased harmonic components and reduce linearity error in measurements. Grootjans et al. [14], Kadambi et al. [15] proposed pseudo-random binary coding sequences to reduce crosstalk and address multi-path interference. Recently, Hamiltonian coding functions were proposed under the proposed coding space theory [6], [7], which could largely improve the depth-imaging accuracy of iToF imaging. Recent advances have connected iToF coding design to compressed sensing theory. Lopez Paredes et al. [9] analyzed the role of coherence in the sensing matrix formed by correlation functions, demonstrating that lower coherence improves signal recoverability in the temporal domain. Yoshida et al. [10] further demonstrated time-compressive sensing using coded exposure patterns. These works highlight the importance of designing coding functions that ensure recoverability with minimal measurements which is a core goal of compressed sensing. In the meantime, different corresponding ToF depth reconstruction algorithms are proposed for depth reconstruction, from the N-step phase-shift algorithm [1], the multiple frequency sinusoid algorithm [16], the zero-mean normalized cross-correlation (ZNCC) [17], to the current deep learning based reconstruction algorithms [18], [5]. Despite significant advancements in depth imaging, current iToF imaging schemes still exhibit relatively limited depth accuracy in low-SNR conditions. To address this challenge, we propose a visual ToF scheme that learns an optimal iToF encoding and incorporates visual information to achieve high-fidelity depth imaging.

B. End-to-End Optimization Framework for Computational Imaging Systems.

End-to-end techniques for simultaneously modifying the imaging process and reconstruction algorithms have gained widespread attention with the development of deep learning and have demonstrated a promising improvement in monocular depth estimation [19], adaptive lidar [20], high dynamic range imaging [21], [22], extended depth-of-field imaging [23], [24], [25], hyperspectral imaging [26], 3D localization microscopy [27], etc. In ToF imaging, Chugunov et al. [18] proposed to jointly learn a microlens amplitude mask pattern and an encoder-decoder network to correct flying pixels in the depth map. Tong et al. [28] proposed a one-stage f^2 -ToF framework to jointly remove MPI, motion blur, and system noise on the raw measurement domain. However, the potential of optimizing the coding function in the iToF imaging scheme is not exploited. In this work, we propose optimizing the coding functions for visual ToF imaging and depth reconstruction. Through establishing a differentiable ToF imaging model with physical constraints, the coding functions and the depth reconstruction neural network can be jointly optimized to realize high depth accuracy.

C. Fisher Information Guided Imaging System Design.

Fisher information measures the amount of information that an observable random variable χ carries about an unknown parameter θ , which could be utilized as guidance for optimizing the design of an imaging system. For snapshot 3D microscopic imaging with high depth accuracy, Shechtman et al. [29] proposed to optimize the pupil phase by maximizing the Fisher information. Chao et al. [30] proposed to locate the single molecule with Fisher information for microscopic imaging under low light. Wu et al. [31] proposed optimizing the phase mask using Fisher information for passive single-view depth estimation. Recent work has bridged Fisher information with compressed sensing by showing that the coherence of the sensing matrix provides an upper bound for information loss [11]. This finding underscores the potential of incorporating information theoretic metrics that directly quantify encoding capacity into the optimization process. While coherence provides an upper bound, Fisher information enables to quantify the information content of measurements under practical noise statistics and specific scene conditions. Despite its proven efficacy in various other imaging domains, Fisher information has not been explored in iToF imaging. In this work, we introduce the Fisher information of the proposed visual ToF imaging scheme to supervise the optimization of coding functions. Besides, since iToF imaging commonly requires multiple measurements for depth extraction, we further introduce a discrimination loss to maximize the difference among the coding functions. Through supervision by the Fisher guidance loss and the discrimination loss (i.e., the discriminative Fisher guidance loss), the coding functions can be effectively optimized to encode sufficient depth information.

D. Visual Information-Guided Depth Image Enhancement.

Visual information-based guidance, such as RGB images, has emerged as an effective enhancement for reconstructing high-quality depth maps from low-quality images. Conventional methods, such as filter-based approaches like bilateral filters [32], guided filters [33], and trilateral filters [34], successfully smooth flat regions and preserve boundaries but struggle with large erroneous areas due to limited local contexts [35], [36]. Optimization-based techniques leveraging models like Markov Random Field (MRF) [37], [38] and Conditional Random Field (CRF) [39], [40] improve upon this but still rely heavily on local contexts, making it challenging to rectify significant errors [41], [42], [43]. Learning-based approaches, such as CNNs designed for depth super-resolution [44], [45], [46], [47], [48], [49], and deformable kernel networks [50], address these limitations, with methods like RGB fusion, effectively mitigating issues like shot noise and sparsity in noisy scenes with strong ambient light [51]. More recent works explore leveraging large pre-trained models like MiDaS [52] and SAM [53] for stronger priors, or even use defective depth data to guide powerful monocular estimators such as Depth Anything [54]. In this paper, we leverage RGB visual information to facilitate the learning and convergence of optimal iToF coding functions, and

to improve absolute depth reconstruction from iToF signals in terms of both accuracy and noise robustness.

III. VISUAL ToF IMAGING FRAMEWORK

A. Overview

The overview of the proposed model is illustrated in Fig. 1. As shown, the visual ToF imaging system is composed of an RGB imaging camera, which captures the visual information, and an iToF imaging system. Based upon this visual ToF imaging system, we propose a discriminative Fisher information-guided visual ToF optimization framework to jointly optimize the coding functions of iToF measurements and the reconstruction network. By differentially modeling the physical iToF imaging process with learnable iToF coding functions, along with a dual-branch visual ToF depth reconstruction network, the proposed method can optimize the entire visual ToF scheme in an end-to-end manner. In this section, we first introduce the proposed differential iToF imaging with practical physics constraints (Section B and C). Based on the imaging model in Section D, we further discuss the iToF imaging noise models and introduce the corresponding discriminative Fisher information guidance for optimizing the iToF coding function.

B. Learnable iToF Imaging Model

We introduce a differentiable iToF imaging model that comprehensively simulates the physical processes from modulated light emission to signal demodulation and measurement, while incorporating realistic physical constraints to closely align with real-world systems. This model enables seamless optimization and supervision through loss functions.

1) *iToF Imaging Process*: Assuming the emitted light is modulated by a periodic signal $M(p)$ and considering the intensity attenuates with the square of the distance, the attenuation is expressed as $F_{\text{falloff}}(z)$. Thus, the reflected signal is:

$$R_i(p, t) = F_{\text{falloff}}(z(p))\alpha(p)M_i\left(t - 2\frac{z(p)}{c}\right) + \beta\alpha(p), \quad (1)$$

where $\alpha(p)$ represents the object's reflectivity, β accounts for environmental light, the time delay $\varphi(p) = 2\frac{z(p)}{c}$ depends on the propagation distance $z(p)$, c is the speed of light, $z(p)$ denotes the depth of light propagation, representing the distance between sensor pixel p and the object.

When p receives the reflected signal $R_i(p, t)$, it is multiplied with the demodulated signal $D_i(t)$ and integrated over time T to obtain the corresponding measurements of $R_i(p, t)$, i.e.,

$$s_i(p) = \int_0^T R_i(p, t)D_i(t)dt, \quad (2)$$

where $s_i(p)$ includes three unknowns: $\alpha(p)$, β , and $z(p)$, requiring at least $N \geq 3$ measurements to determine $z(p)$. By modeling $M_i(t)$ and $D_i(t)$ as differential functions with learnable parameters, the iToF imaging process can be modeled as a differentiable iToF physical imaging framework.

C. iToF Imaging Physical Constraints

To ensure the learned coding functions are physically realizable, our optimization process is guided by several hardware-aware constraints. First, we incorporate the bandwidth constraint by modeling $M(t)$ and $D(t)$ as truncated sums of n sinusoidal components (3), which guarantees hardware-feasible spectral support.

$$\begin{aligned} M_i(t) &= \sum_{i=1}^n a_i^m \sin(2\pi f_0 t + \varphi_i^m) + b_m, \\ D_i(t) &= \sum_{i=1}^n a_i^d \sin(2\pi f_0 t + \varphi_i^d) + b_d, \end{aligned} \quad (3)$$

where f_0 denotes the fundamental frequency and $f_{\max} = n f_0$. a_i^m (a_i^d) and φ_i^m (φ_i^d) represent the amplitude and phase of the i -th harmonic component $f_i = i f_0$ for $M(t)$ ($D(t)$), and b_m and b_d are the corresponding DC offsets. In addition, we enforce practical physical constraints, including a bounded illumination energy budget, non-negative modulation/demodulation signals, and unit-normalized demodulation amplitude, to ensure hardware-feasible coding functions (see [8] for details).

D. Fisher Information of iToF Imaging System

Fisher information, a metric for the information about unknown parameters from observations, guides our optimization of iToF systems by seeking measurement schemes with the highest information content.

1) *iToF Imaging System Noise Modeling*: To derive the Fisher information of an iToF measurement scheme, we investigate the signal and noise distribution of the measurement scheme first. Commonly, the measurements of the iToF system X_i on scene point p can be divided as

$$X_i(p) = S_i(p) + N_{\text{dark}} + N_{\text{readout}}, \quad (4)$$

where $S_i(p)$ is the depth-dependent optical signal that follows the Poisson distribution, and $s_i(p)$ is the expectation of $S_i(p)$ that depends on depth, i.e. $s_i(p) = E[S_i(p)]$. $N_{\text{dark}} \sim \mathcal{P}(\lambda_d)$ is the dark current noise with Poisson distribution, and λ_d is the expectation of the dark current noise. $N_{\text{readout}} \sim \mathcal{N}(0, \sigma_r^2)$ is Gaussian distributed readout noise, where σ_r is the standard deviation of the readout noise. We approximate the Poisson noise with additive Gaussian noise of various parameters, i.e.,

$$\begin{aligned} X_i(p) &\sim \mathcal{N}(\mu_i(p), \sigma_i^2(p)), \\ u_i(p) &= s_i(p) + \lambda_d, \quad \sigma_i(p) = \sqrt{s_i(p) + \lambda_d + \sigma_r^2}. \end{aligned} \quad (5)$$

2) *Fisher Information Guidance*: Note that here, the dark noise and readout noise are independent of depth $z(p)$ and depend only on the status of the detecting sensor. Typically, for each scene point in iToF imaging, multiple measurements $X(p) = [x_1(p), x_2(p), \dots, x_N(p)]$ are captured with different combinations of modulation and demodulation functions. N is the number of measurements of each scene point, and we could derive the Fisher information of the observed variable with respect to depth, i.e.

$$\mathbb{I}(X(p); z(p)) = \sum_i \left[\frac{1}{2\sigma_i^4(p)} + \frac{1}{\sigma_i^2(p)} \right] \left[\frac{\partial s_i(p)}{\partial z} \right]^2. \quad (6)$$

IV. DUAL-BRANCH VISUAL TOF DEPTH RECONSTRUCTION NETWORK

Utilizing the visual ToF imaging scheme, we capture a series of iToF measurements represented as $I_{\text{iToF}} = [I_1, I_2, \dots, I_N]^T$ ($N \geq 3$) and RGB image represented as I_{RGB} . To achieve depth reconstruction from the iToF measurements and RGB image, we propose a dual-branch visual ToF depth reconstruction network. As illustrated in Fig. 2, the network consists principally of two main branches: one branch serves as the iToF depth reconstruction network, which extracts depth information from the iToF measurements, and the other branch serves as an RGB edge-guided depth enhancement branch, aimed at introducing spatial information for both the efficient learning of iToF coding functions and high precision depth reconstruction.

A. iToF Depth Reconstruction Network

1) *Spatial-Structure Extraction Branch (SEB)*: We propose a feature pyramid network to incorporate multi-scale spatial structural information and reconstruct the depth map with a sparsity prior. Dense blocks are used for the feature encoding. Three feature-extraction blocks were used in the encoder, and the extracted multi-scale features were fused to produce features at varying scales. The decoder is built using a coarse-to-fine approach. To create an enlarged-resolution depth map, we first construct a decoder that predicts the smallest-scale depth map. We then up-sample using bilinear interpolation and integrate with higher-scale fusion features. The ultimate depth map is obtained by repeating this procedure. In this sense, the global information in the small-scale depth map is continuously updated to produce more detailed large-scale depth maps.

2) *Pixel-Wise Depth Regression Branch (PDRB)*: Through single-point direct mapping, the obtained depth map from the iToF lower branch can retain the detailed features of the scene. The pixel-wise depth regression branch serves as a simulator of the conventional depth extraction method for iToF imaging, i.e., it extracts the depth of each pixel from the corresponding measurements and ensures fast convergence in learning the coding functions. Note that, as shown in Fig. 2, for sufficient information fusion between the two iToF branches, we introduce feature concatenation from the SEB branch to the PDRB.

B. RGB Edge-guided Depth Enhancement Network

We present the RGB edge-guided depth enhancement network designed to introduce more spatial details to the iToF imaging and reconstruction branch and further improve the reconstructed depth from the iToF branch. The overall structure of RGB edge-guided depth enhancement network is illustrated in Fig. 2, which comprises two primary modules: the bimodal edge-focused feature extraction module (BEM) and the iToF edge fusion module (TEM).

1) *Bimodal Edge-Focused Feature Extraction Module (BEM)*: To fully capture spatial details, we constructed a bimodal feature extraction module that leverages edge information

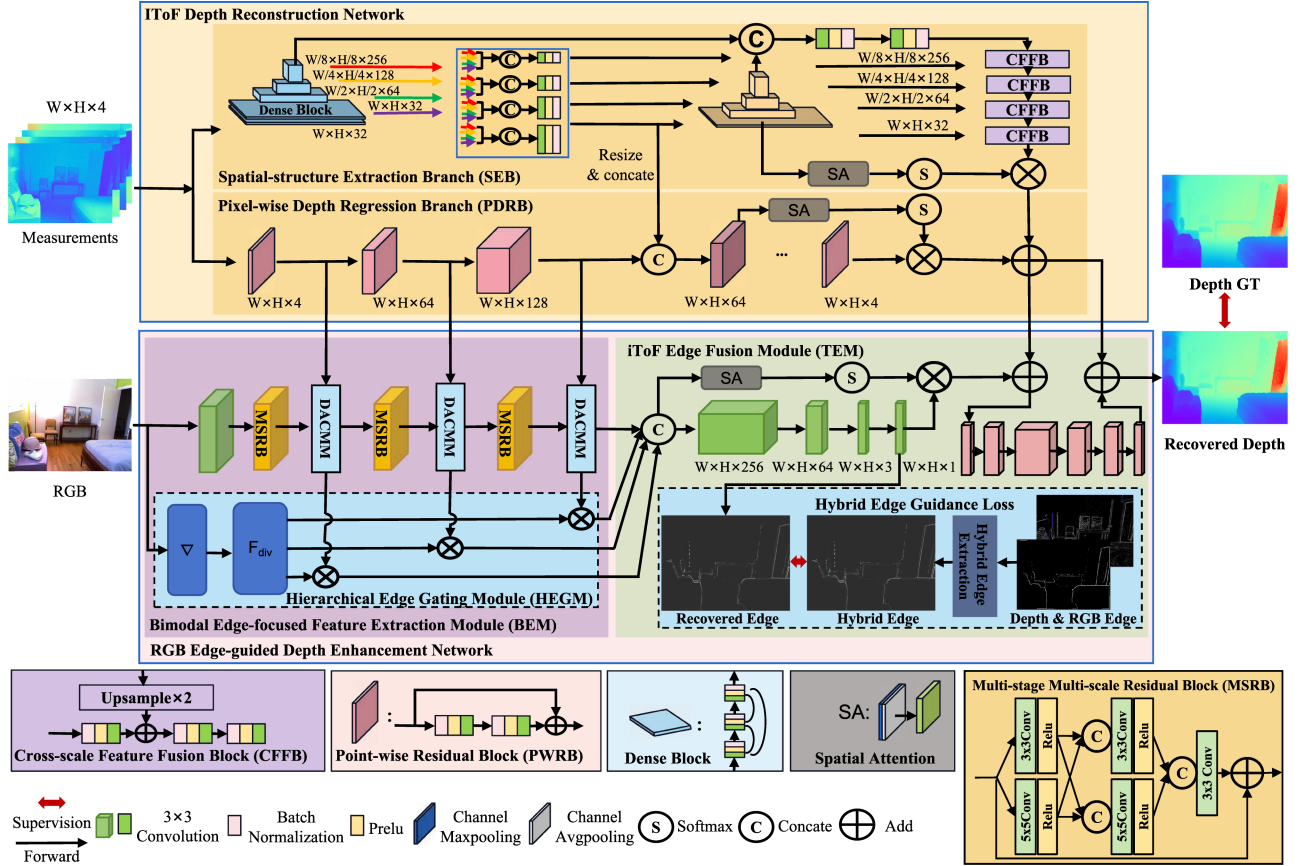


Fig. 2. An overview of our dual-branch visual ToF depth reconstruction network, different color squares represent different network modules.

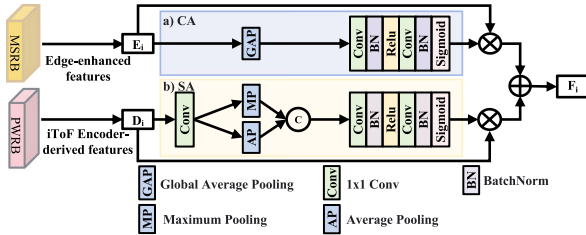


Fig. 3. Network structure of the proposed DACMM module. E_i denotes features from the MSRB module, and D_i denotes features from encoders of the iToF Depth Reconstruction Network.

from both the RGB image and the iToF features. Specifically, we develop a backbone that employs multi-stage, multi-scale residual blocks (MSRB) [55] as shown in Fig. 2, eliminating the downsampling operation to perform all processing at the same spatial scale, thereby enhancing the extraction of sufficient spatial information. Building upon the backbone, we further incorporate features from the iToF depth reconstruction branch using the proposed dual-alternative combined multimodal module (DACMM), as illustrated in Fig. 3. To focus exclusively on edge information extraction for edge guidance, we propose a Hierarchical Edge Gating Module (HEGM) that enables the network to prioritize feature extraction around edges, as depicted in the BEM module in Fig. 2.

Dual-alternative Combined Multimodal Module (DACMM):

The integration of iToF features minimizes artifact generation and directs depth-edge extraction from RGB images, which is particularly advantageous for enhancing the iToF-reconstructed depth image. To effectively fuse the iToF and vision edge features, we introduce the DACMM module, comprising two attention branches, as illustrated in Fig. 3. The spatial attention (SA) mechanism enhances the critical spatial information of the iToF features from cascaded residual blocks to guide the BEM to generate more iToF-consistent edges, while the channel attention (CA) mechanism selectively focuses on the relevant RGB feature channels that contain rich boundary and texture information.

As shown in Fig. 3, the attention-weighted fusion feature is generated through a dual-branch process. For the RGB branch, the feature E_i undergoes channel attention computation by first applying global average pooling (GAP), followed by two sequential 1×1 convolutional layers with batch normalization and ReLU activation, and finally a sigmoid activation to produce the channel attention weight $A(E_i)$. This emphasizes channel-wise dependencies within the RGB features. For the iToF branch, the feature D_i is processed to derive spatial attention weights: after a 1×1 convolution and ReLU activation, maximum pooling (MP) and average pooling (AP) are applied to D_i , concatenated along the channel dimension, and then passed through a 1×1 convolution, batch normalization, and sigmoid activation to

obtain the spatial attention weight $A(D_i)$. Finally, the fused feature E_{i+1} is computed by element-wise multiplication of D_i with $A(D_i)$ and E_i with $A(E_i)$, followed by summation of the weighted features. This attention-guided fusion enables adaptive integration of edge-enhanced RGB cues and spatially refined iToF features, which are then propagated to subsequent MSRB blocks and seamlessly merged into the iToF depth reconstruction branch.

Hierarchical Edge Gating Module (HEGM): To efficiently extract sufficient edge information, we propose the hierarchical edge gating module (HEGM), which uses the RGB edge to gate feature extraction on fused bimodal features from the DACMM module. Specifically, we first calculate the gradient of the RGB image, followed by its divergence, to generate the edge-gating mask, i.e.,

$$I_{\text{Edge}} = F_{\text{div}}(\nabla_x I, \nabla_y I), \quad (7)$$

where x denotes horizontal dimension, y denotes vertical dimension, ∇ denotes the gradient operator, and $F_{\text{div}}(\cdot)$ denotes the divergence operator, respectively. To implement gating, we propose multiplying this mask by the attention-fused features from the DACMM modules at various backbone depths. We then employ concatenation to fuse three levels of gated features, ensuring sufficient edge information is preserved.

2) iToF Edge Fusion Module (TEM): Based on the edge features extracted by the BEM module, we developed the TEM module to extract depth edges while excluding texture edges. This design enables the provision of residuals that further improve the depth information derived from the iToF reconstructed depths. Consequently, the TEM is structured in two primary phases: the initial phase is dedicated to extracting depth edges, which we denote as Recovered Edge R , defined as the intermediate edge output (dimension $W \times H \times 1$) generated by the edge fusion module as shown in Fig. 2, with the proposed guidance supervision of the hybrid edge guidance loss. By omitting texture edges, we have implemented a streamlined residual module that learns the residuals surrounding the edges, thereby facilitating the refinement of depth information reconstructed from the iToF.

As depicted in Fig. 2, in the first stage, we introduce an attention-based module, wherein one branch reduces the feature channel dimension through the channel attention module, followed by a subsequent softmax module. To refine the learning process, we propose a hybrid edge guidance loss to supervise an additional branch that learns a depth edge mask, effectively guiding the network to focus on depth discontinuities rather than texture-induced edges. The outputs of these two branches are then multiplied to achieve a more sophisticated extraction of depth-discontinuity edge features.

Hybrid Edge Detection: Limited by the quality of the depth image, the edges it produces are relatively rough, whereas those from the RGB image are sharper but exhibit more texture-induced artifacts. Thus, we propose to learn a mask specifically for edges caused by depth discontinuities, using supervision based solely on the depth edge mask, as illustrated in Fig. 4. As in (8) and (9), we first utilize a soft-edge technique [56] to generate the edges of both RGB and iToF images. Compared with conventional edge detectors, this method preserves more

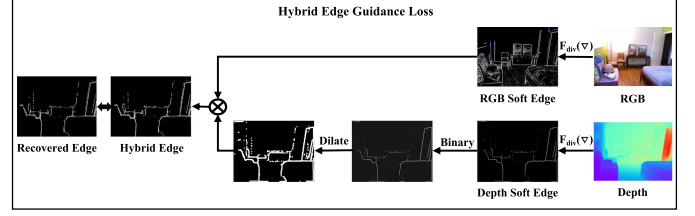


Fig. 4. The pipeline of the hybrid edge guidance loss.

precise edge information by employing the following formula to derive the corresponding soft-edge I_{Edge} from the RGB and depth ground truth, thereby effectively capturing the variations in the gradient domain.

$$I_{\text{RGB}}^{\text{Edge}} = F_{\text{div}}(\nabla_x I_{\text{RGB}}, \nabla_y I_{\text{RGB}}), \quad (8)$$

$$I_{\text{iToF}}^{\text{Edge}} = F_{\text{div}}(\nabla_x I_{\text{iToF}}, \nabla_y I_{\text{iToF}}), \quad (9)$$

where I_{RGB} and I_{iToF} denote the RGB image and depth image respectively. $I_{\text{RGB}}^{\text{Edge}}$ and $I_{\text{iToF}}^{\text{Edge}}$ is the extracted edge from RGB and depth image.

With the extracted edge from the depth and RGB image, we then generate a fused edge map F_{Edge} for hybrid edge guidance supervision by extracting edges from the depth map, creating a binary mask, and applying dilated operations to filter and refine the RGB edges, as illustrated in (10),

$$F_{\text{Edge}} = I_{\text{RGB}}^{\text{Edge}} * D \left\{ O \left\{ I_{\text{iToF}}^{\text{Edge}} \right\} \right\}, \quad (10)$$

where $O\{\cdot\}$ represents the binary operations which is denoted as $O = \begin{cases} 1, & x \geq C \\ 0, & \text{otherwise} \end{cases}$ where x is the pixel intensity in the image and C is the threshold value to determine the binarization cutoff. $D\{\cdot\}$ represents the dilated operations. The dilation of image I by structuring element S is given by: $I \oplus S = \{z \mid (\hat{S})_z \cap I \neq \emptyset\}$ where z represents a translation vector. This hybrid edge detect process not only preserves the sharpness of RGB edges but also effectively removes inconsistent edges caused by artifacts and textures, resulting in a more accurate and detailed edge representation utilized for hybrid edge guidance supervision within the TEM.

C. Loss Function

In this work, we use hybrid edge guidance loss, the discriminative Fisher loss, comprising Fisher information loss and the multiple measurement difference loss function, and the depth fidelity loss function, to jointly train the coding function and the depth reconstruction network. The overall loss function formula of the system is constructed as the weighted combination of hybrid edge guidance loss, the discriminative Fisher guidance loss, and fidelity loss, i.e.

$$L = L_{\text{Fidelity}} + \lambda_1 L_{\text{dFisher}} + \lambda_2 L_{\text{Edge}}, \quad (11)$$

where L_{Fidelity} is the depth fidelity loss, L_{dFisher} is the summation of fisher information guidance loss and multi measurement discriminative loss, i.e., $L_{\text{Fisher}} = L_{\text{Fisher}} + \lambda_3 L_{\text{Dis}}$, L_{Edge} is the

edge loss, and λ_1 , λ_2 , and λ_3 are the hyperparameters of network training.

Fisher Guidance Loss: Given the fisher information of measurements $s_i(p)$ on depth $z(p)$, we can derive a fisher guidance loss as

$$L_{\text{fisher}} = - \sum_p \mathcal{I}_{\text{fisher}}(p). \quad (12)$$

Fidelity Loss: To enforce the depth reconstruction fidelity, we proposed to minimize the mean absolute error (MAE) between the depth of the reconstruction result z^* and the target depth z_{gt} , i.e.,

$$L_{\text{Fidelity}} = \frac{1}{N} \sum_p \|z^*(p) - z_{\text{gt}}(p)\|_1. \quad (13)$$

Multi-measurement Discriminative Loss: To further improve the efficiency of different coding functions, and enforce the difference between different coding functions, here we adopt the Manhattan distance as the discriminative loss, i.e.

$$L_{\text{Dis}} = - \sum_p \sum_{0 \leq i \leq N, 0 \leq j \leq N} \|c_i(p) - c_j(p)\|_1, \quad (14)$$

where c is the correlation function, which is the product of the modulation and demodulation functions, i.e., $c_i(t) = M_i(t)D_i(t)$.

Hybrid Edge Guidance Loss: We propose a fusion edge supervision loss function to further improve the edge network learning's consistency with the deep modality and accuracy of the edge texture, i.e.

$$\mathcal{L}_{\text{Edge}} = \|R(I_{\text{RGB}}) - F_{\text{Edge}}\|_1, \quad (15)$$

where R denotes the reconstruction edge of the RGB edge-guided depth enhancement network and I_{RGB} denotes the RGB image.

V. EXPERIMENTAL RESULTS

A. Datasets

We utilize the NYU-V2 dataset [57] to train our visual ToF imaging system, which integrates both the temporal coding function and the Visual ToF depth reconstruction network. This dataset comprises video sequences from 464 indoor scenes, featuring a total of 1,449 densely labeled pairs of aligned RGB and depth images. The albedo map of the scene can be generated using the RGB of the scene and corresponding depth map after intrinsic decomposition [58]. Following the methodologies outlined in [59] and [50], we sample 1,000 RGB-D images for training, reserving the remaining 449 pairs for testing.

This study employs two additional datasets from real RGB-D assemblies: the SUN RGB-D dataset [60] and the simulated 4D Light Field datasets [61]. The SUN RGB-D dataset includes 10335 depth maps paired with aligned RGB images across 47 indoor scenes and 800 object categories, captured by diverse depth cameras including Intel Realsense, Asus Xtion, and Microsoft Kinect V1 and V2. Similarly, the 4D Light Field dataset, generated using Blender's internal renderer, features $9 \times 9 \times 512 \times 512 \times 3$ light field RGB images and 16-bit depth maps. Both datasets provide RGB images and corresponding

depth maps that satisfy our experimental requirements. For model generalization verification, we select 200 RGB-D image pairs from the SUN RGB-D dataset and 14 pairs from the 4D Light Field dataset.

Focusing on depth sensing, this paper evaluates the proposed method using four commonly employed metrics [62], such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Absolute Relative Difference (Abs. Rel) and Squared Relative Difference (Sq. Rel).

B. Training Details

In addition to the readout noise and dark noise of the image sensor, which we assume as constant and choose to be 20 electrons in our experiment. The SNR of the iToF measurements is mainly determined by the incident light source and the ambient light powers. In a typical iToF imaging scheme operating under high ambient noise conditions, a modulated laser source with a power of 30 to 100 mW scans the beam onto the target object, which is exposed to substantial ambient light levels ranging from 5000 to 10,000 lx. The reflected signal is captured by the detector, demodulated, and integrated over 0.1 ms. Under these conditions, the system accumulates approximately 10,000 to 20,000 photons from the reflected signal and 2,000 to 6,000 photons from ambient light, accounting for overall system attenuation. Thus, to train the proposed network to achieve depth reconstruction under different noise levels, we set E_{incident} , which is the number of incident light photons, and E_{ambient} , which is the number of ambient light photons to (20000, 6000), (14000, 6000), and (10000, 6000) to simulate physical experiments and cover a wide range of noise levels from low to high, i.e., from 22 dB to 16 dB. Besides, given the physical implementation feasibility, we set the bandwidth limit based on practical conditions, i.e., 50 MHz as the fundamental frequency and 250 MHz as the bandwidth limit.

As for the training strategy, to train the proposed network and deal with different noise levels simultaneously, we adopt the incremental training strategy [64]. The network is trained on input data with different noise levels, ranging from small to large, every 10 epochs. When traversing all noise levels, samples with random noise levels are generated and fed to the network to facilitate its convergence. We adopt ADAM [65] as the optimizer, with an initial learning rate of 0.01. The learning rate decays linearly at a rate of 0.7 every 10 epochs. λ_1 , λ_2 and λ_3 were chosen empirically to be 1×10^{-4} , 1 and 1×10^{-1} initially and decayed to 1×10^{-5} after 10 epochs for λ_1 . Kaiming initialization [66] is utilized for the learnable modulation and demodulation functions. We run the experiments on the PyTorch platform [67] using an NVIDIA GeForce RTX 3090 GPU.

C. Evaluation on the Synthesized Data

To demonstrate the superior performance, we compare our method with existing iToF imaging methods.

1) *Comparison With Existing iToF Imaging Methods:* We compare our method with various state-of-the-art iToF imaging methods, including the conventional sinusoid and square coding functions with the conventional phase shift (PS) algorithm [1],

TABLE I
OVERALL QUANTITATIVE COMPARISONS WITH EXISTING METHOD ON NYU-V2 DATASET

Algorithms	SNR = 22 dB				SNR = 19 dB				SNR = 16 dB			
	MAE↓	RMSE↓	Abs. Rel↓	Sq. Rel↓	MAE↓	RMSE↓	Abs. Rel↓	Sq. Rel↓	MAE↓	RMSE↓	Abs. Rel↓	Sq. Rel↓
Sinusoid + PS	183.85	303.05	0.12	18.62	247.40	387.68	0.17	31.27	318.00	474.03	0.22	49.03
Square + PS	115.23	194.50	0.07	7.70	161.37	270.32	0.10	14.78	216.59	349.00	0.14	24.93
Hamilton + ZNCC	100.26	242.53	0.06	11.49	151.03	324.84	0.09	20.59	210.34	403.17	0.13	32.31
Sinusoid + DeepToF	57.91	78.54	0.0452	1.509	107.23	128.84	0.0883	4.525	193.26	221.28	0.1763	15.799
C2F-DenseCRF	23.66	27.42	0.0125	0.1803	25.77	31.88	0.0119	0.2010	32.52	42.33	0.0195	0.2371
Sinusoid + Base	17.39	25.12	0.0133	0.1458	25.10	32.06	0.0162	0.1970	31.72	35.38	0.0163	0.2198
Base	12.86	22.07	0.0118	0.1181	19.50	28.20	0.0156	0.1966	22.11	32.54	0.0155	0.2158
7phase	9.79	18.44	0.0095	0.0894	16.81	25.34	0.0127	0.1492	21.34	31.50	0.0159	0.2177
Our method	9.36	16.50	0.0090	0.0762	11.8	18.85	0.0091	0.1004	17.53	28.38	0.0146	0.2112

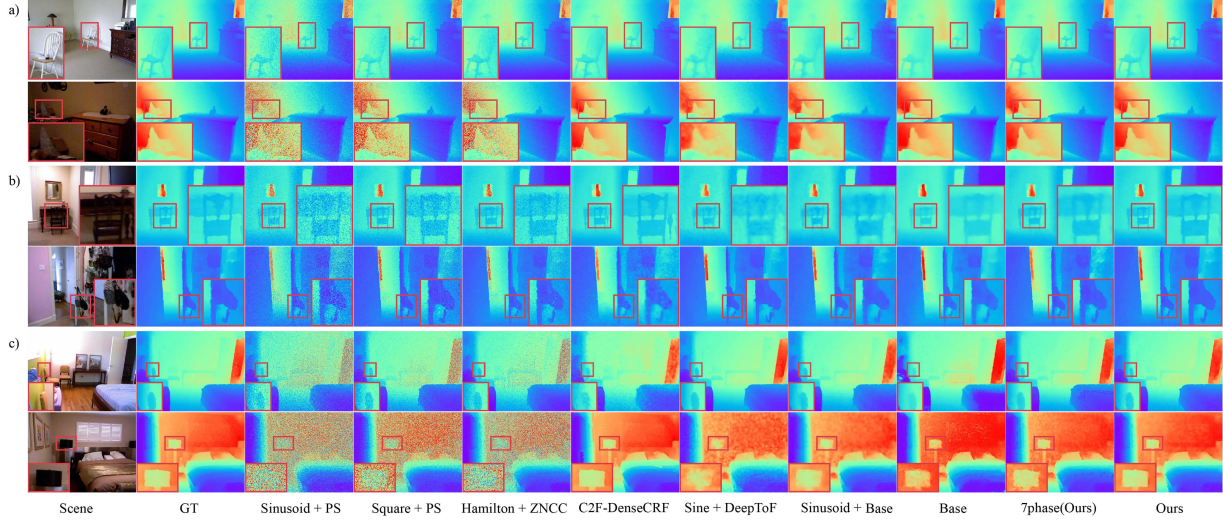


Fig. 5. Visual comparison with the state-of-the-art iToF methods, i.e. Sine/Square+PS algorithm [1], Hamilton [7], C2F-DenseCRF [63], and Sine+DeepToF [5], our Base method [8] and its “7-phase” variation under different noise levels: (a) 22 dB, (b) 19 dB, and (c) 16 dB.

TABLE II
COMPARISON WITH TOF-ONLY, RGB-ENHANCED TOF, AND GENERIC RGB-D FUSION METHODS ON THE NYU-V2 DATASET

Algorithms	SNR = 22 dB				SNR = 19 dB				SNR = 16 dB			
	MAE↓	RMSE↓	Abs. Rel↓	Sq. Rel↓	MAE↓	RMSE↓	Abs. Rel↓	Sq. Rel↓	MAE↓	RMSE↓	Abs. Rel↓	Sq. Rel↓
DeepToF (iToF + RGB) [68]	34.98	53.32	0.0300	0.9001	43.25	63.07	0.0359	1.1626	82.49	121.58	0.0634	2.9078
Wild-ToFu [51]	32.13	52.78	0.0243	0.7862	37.26	58.63	0.0278	0.8984	58.59	82.86	0.0410	1.6659
iToF + MiDaS [52]	30.53	43.32	0.0221	0.4261	46.79	63.62	0.0336	0.9047	79.88	104.77	0.0583	2.4951
iToF + SAM [53]	23.52	31.81	0.0169	0.2267	35.13	46.37	0.0254	0.4992	55.61	71.59	0.0422	1.2994
PDA (baseSine) [54]	29.88	68.17	0.0212	1.1772	35.64	75.96	0.0255	1.4292	46.77	90.53	0.0334	1.9494
PDA (baseLearned) [54]	26.92	63.37	0.0190	1.0364	29.92	67.18	0.0210	1.1415	37.77	76.65	0.0268	1.4281
Ours (MiDaS [52])	15.28	22.36	0.0118	0.1180	24.72	35.26	0.0219	0.4043	53.52	70.03	0.0334	1.2856
Ours (SAM [53])	11.94	17.39	0.0094	0.0721	15.32	23.01	0.0123	0.1365	22.83	35.63	0.0185	0.3480
Ours	9.36	16.50	0.0090	0.0762	11.8	18.85	0.0091	0.1004	17.53	28.38	0.0146	0.2112

the Hamilton iToF imaging method [7], C2F-DenseCRF [63], the DeepToF method [5], the approach introduced in our prior work [8], denoted as “Base”, and its variation. Specifically, the method “Sinusoid + Base” is compared with the Base, which uses sine coding functions and the reconstruction network from the Base method. Further, since our method uses three additional channels of RGB images, we also compare it with the “7-phase” variations of the Base method, ensuring a fair comparison with the same number of captured measurements and a 7-step phase shift. Without loss of generality, our method utilizes one modulation function and four demodulation functions. The results of three different noisy scenarios are shown in Table I, ranging from low noise level to high. Our method achieves the lowest MAE for reconstruction depth across different noise levels. The qualitative comparisons are presented in Fig. 5. The

proposed method outperforms existing iToF imaging techniques across various noise scenarios, demonstrating significant advantages in reduced noise, fewer errors, and enhanced edge detection.

2) *Comparison With RGBD Imaging Methods:* We further conduct a comprehensive comparison with several representative RGB-guided and RGB-D fusion methods. As shown in Table II, our method significantly outperforms both early-fusion baselines, such as Wild-ToFu [51] and an RGB-extended DeepToF [51], which struggle to effectively merge the two modalities. We also compare against strategies that integrate large pre-trained models. While integrating lightweight SAM [53] or MiDaS [52] into our framework (Ours (SAM / MiDaS)) improves performance over the baselines, our proposed end-to-end model still achieves superior results, highlighting

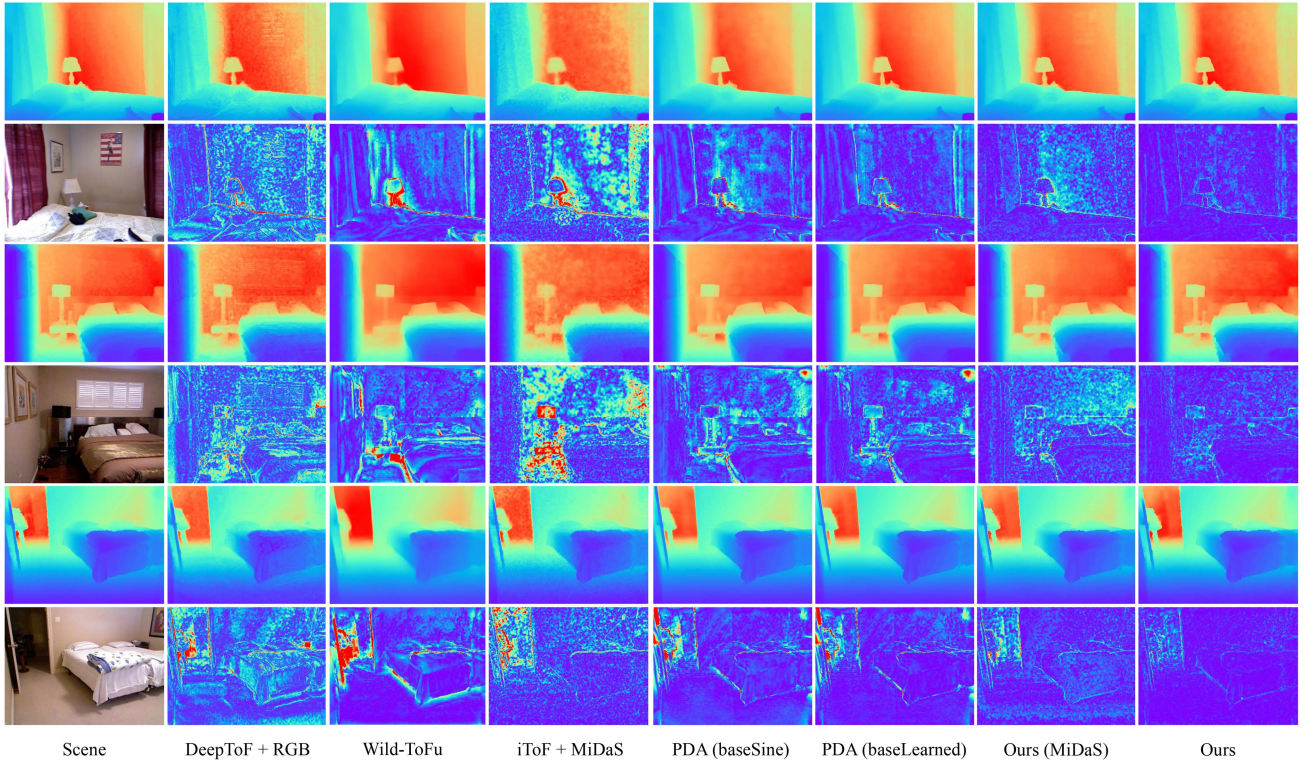


Fig. 6. Visual comparison with the state-of-the-art RGBD methods, i.e., DeepToF [68] + RGB, WildToFu [51], PriorDA [54], Our + MiDaS [52] and Ours + SAM [69] under noise level 22 dB.

TABLE III
QUANTITATIVE COMPARISONS WITH DIFFERENT CODING FUNCTIONS ON THE NYU-V2 DATASET

Algorithms	SNR = 22 dB				SNR = 19 dB				SNR = 16 dB			
	MAE↓	RMSE↓	Abs. Rel↓	Sq. Rel↓	MAE↓	RMSE↓	Abs. Rel↓	Sq. Rel↓	MAE↓	RMSE↓	Abs. Rel↓	Sq. Rel↓
Sinusoid	28.85	115.47	0.07	4.12	72.51	264.75	0.15	15.11	135.38	465.08	0.28	45.67
Square	24.11	95.50	0.06	2.90	34.56	134.41	0.07	4.32	64.41	240.28	0.12	12.01
Dual-freq Sinusoid	21.93	38.14	0.0161	0.3968	36.17	56.19	0.0237	0.8846	40.21	68.63	0.0290	1.1189
Hamiltonian	13.18	23.47	0.0137	0.1398	22.77	28.84	0.0174	0.2010	26.73	33.66	0.0204	0.2723
Base	12.86	22.07	0.0118	0.1181	19.50	28.20	0.0156	0.1966	22.11	32.54	0.0155	0.2158
Our method	9.36	16.50	0.0090	0.0762	11.8	18.85	0.0091	0.1004	17.53	28.38	0.0146	0.2112

the benefits of our dedicated edge-aware fusion mechanism. Furthermore, we evaluate Prior Depth Anything (PDA) [54], a state-of-the-art monocular depth model guided by depth priors. Although PDA produces visually plausible results, it suffers from over-smoothing at object boundaries and can be misled by noisy priors, leading to higher numerical errors compared to our method. Across all comparisons, our full model consistently yields the best performance in key metrics like MAE and RMSE, producing sharper edges and more accurate structural details, which is further validated by our qualitative results in Fig. 6.

3) *Comparison of Different Coding Functions:* To further demonstrate the superiority of the proposed learned coding functions, we compare our method with other coding functions, including sinusoid, square, and dual frequency sinusoid (50 MHz and 200 MHz), as well as the practical Hamilton functions [7], using the same reconstruction neural network, specifically the proposed dual-branch visual ToF depth reconstruction network. The quantitative and qualitative results are presented

in Fig. 7 and Table III. As compared, the learned coding function achieves the best performance, with a significant improvement, especially at high noise levels, further demonstrating the noise robustness of the learned coding functions. The bottom line illustrates the reconstructed depth value along the green line marked in the scene image. It is evident that the reconstruction result of our method achieves the closest performance to the ground-truth depth value.

D. The Influence of Visual Information

1) *Learned Coding Functions With and Without the Visual Information:* To compare changes in coding functions and Fisher information after introducing visual information, we trained the baseline method [8] (without visual information) and our method (with visual information) across three noise levels. As shown in Fig. 8(b), we observe that the coding functions guided by visual information achieve better convergence, with a larger modulation range (i.e., the waveform spans a greater portion of

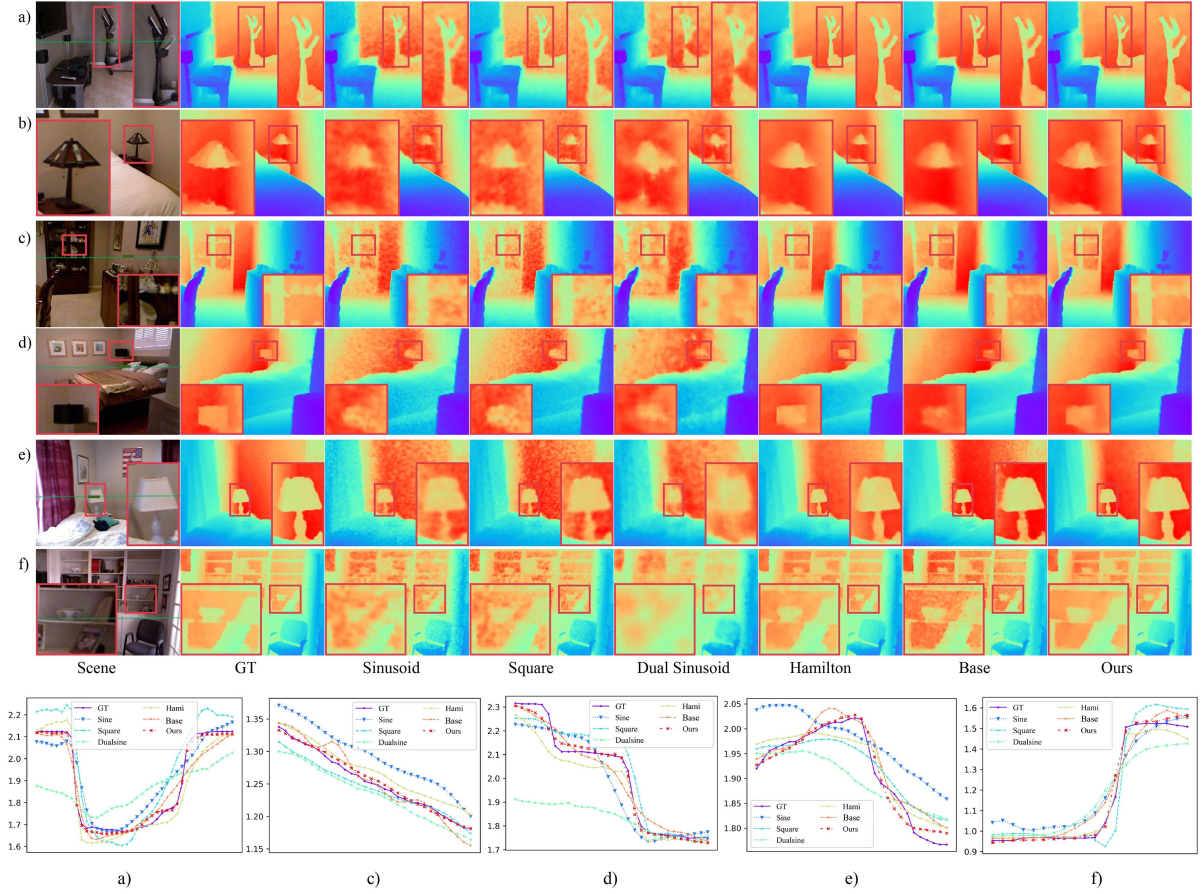


Fig. 7. Depth imaging performance comparisons with different coding functions with the proposed dual-branch visual ToF depth reconstruction network under different noise levels: a-b: 22 dB, c-d: 19 dB, and e-f: 16 dB. The bottom line shows the corresponding reconstructed depth value along the green line marked in the scene images.

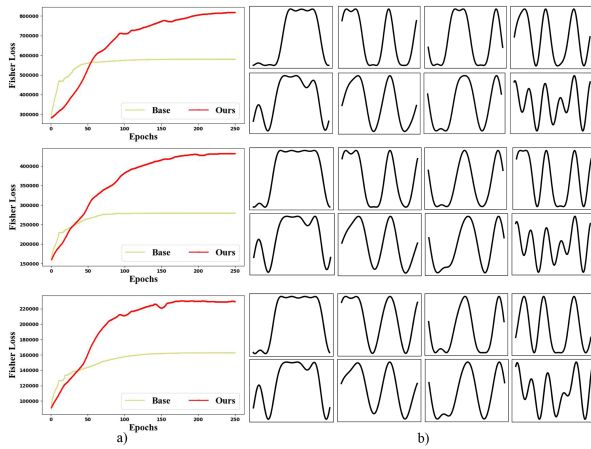


Fig. 8. Comparison of the Fisher information a) and corresponding coding function b) of the Base method [8] and our method in three noise levels, which are 22, 19, and 16 dB from upper to bottom. The first and second lines in each group in b) are the coding function waveforms for our method and the Base method, respectively.

the 0-1 interval) across different depths and demonstrate robustness under various noise conditions. In contrast, without visual guidance, the coding functions converge earlier and exhibit a

TABLE IV
COMPARISON OF FISHER INFORMATION OF DIFFERENT CODING FUNCTIONS

Algorithms	Sinusoid	Square	Hamiltonian	Base	Our
Fisher information	1.1e+5	4.5e+5	6.3e+5	7.7e+5	9.2e+5

narrower modulation range across depths, suggesting that the lack of visual information may limit the exploration of more optimal solutions. This further underscores the effectiveness of incorporating visual information to enhance the learning of coding functions with improved depth information encoding capabilities.

2) *Changes in the Fisher Information Caused by the Introduction of Visual Information:* According to (6), we calculated the Fisher information of the measurement values obtained from the learnable coding functions proposed in this paper, as well as other existing coding functions under the noise levels of (20000, 6000). The comparison results are shown in the Table IV. Our proposed learnable coding function achieves the highest Fisher information, which indicates that the measurements obtained by our coding functions can encode more depth information.

Furthermore, as shown in Fig. 8(a), the Fisher loss of the Base method [8] converges quickly to a low Fisher information

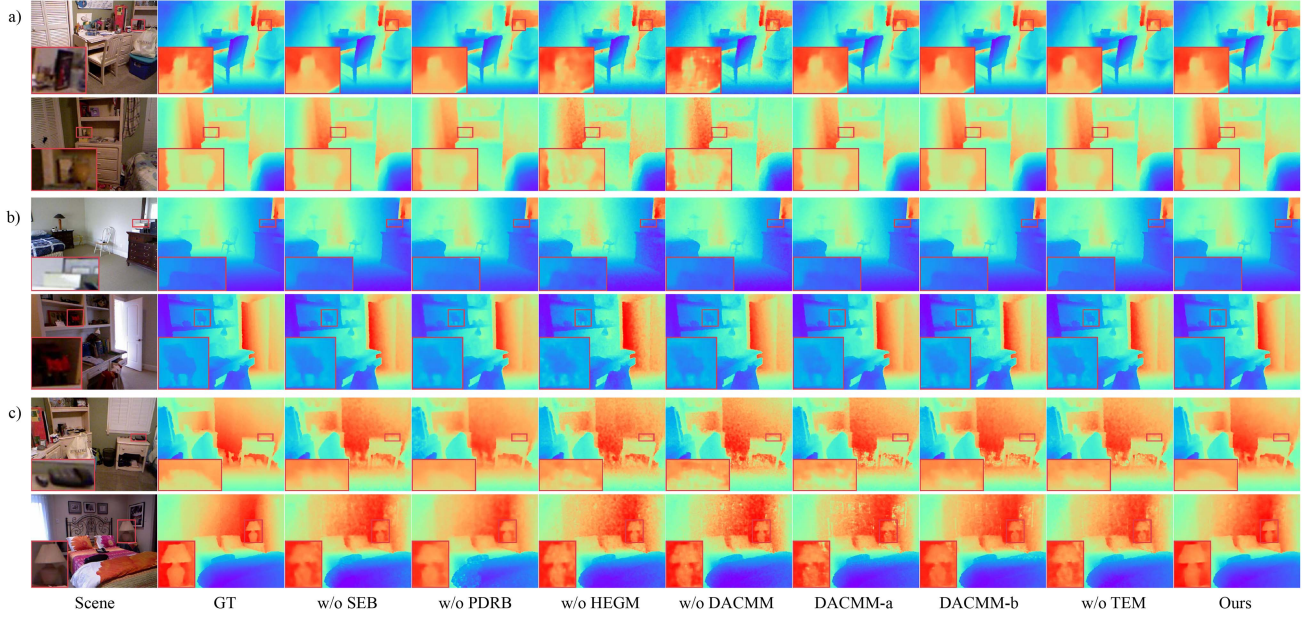


Fig. 9. Ablation study of different network modules under different noise levels, i.e., (a) 22 dB, (b) 19 dB, (c) 16 dB.

TABLE V
ABLATION STUDY ON THE EFFECT OF THE NETWORK MODULES

SEB	PDRB	HEGM	DACMM	TEM	SNR = 22 dB		SNR = 19 dB		SNR = 16 dB	
					MAE↓	RMSE↓	MAE↓	RMSE↓	MAE↓	RMSE↓
×	✓	✓	✓	✓	12.57	19.79	19.43	28.66	42.57	57.34
✓	×	✓	✓	✓	18.30	25.45	27.24	34.53	72.98	91.41
✓	✓	×	✓	✓	15.27	21.26	18.42	25.12	33.43	45.38
✓	✓	✓	×	✓	14.36	25.99	19.05	32.51	35.18	46.24
✓	✓	✓	×	✓	13.03	23.95	18.08	32.76	31.71	53.06
✓	✓	✓	×	✓	12.94	22.76	18.96	32.07	29.56	49.77
✓	✓	✓	×	×	13.76	20.55	20.33	28.26	35.08	47.84
✓	✓	✓	✓	✓	9.36	16.50	11.82	18.85	17.53	28.38

value, whereas our proposed method continues to increase and ultimately converges at a much higher value. This indicates that, compared with our method, the Base method struggles to optimize the coding function in noisy environments due to the difficulty in distinguishing depth changes influenced by substantial noise. As a result, the Base method typically converges earlier and at a lower level. In contrast, by incorporating visual information, our method leverages additional prior knowledge of depth variations, preventing the learning process from becoming trapped in local optima. This enables the development of superior coding functions. These findings further validate the effectiveness of the proposed visual ToF imaging scheme.

3) *Changes in the Performance of the Reconstruction Caused by the Introduction of Visual Information:* Our method incorporates visual information to enhance the performance of iToF imaging, particularly in preserving edge textures. As shown in Table I, our method achieves a 3-8 mm reduction in MAE compared to the Base method [8], demonstrating improved overall and local detail reconstruction. Qualitative comparisons, presented in Fig. 5, further illustrate the advantages of our approach. In terms of reconstruction visualization, our method not only provides more robust depth reconstruction under low signal-to-noise ratio conditions but also excels at capturing finer

details, such as school bag straps and seat backs, compared to the Base method.

E. Ablation Study

In this section, we present the ablation study to evaluate the different components of the proposed algorithm, including the proposed network modules and the loss functions.

1) *Effect of the different network modules:* To evaluate the effectiveness of the proposed SEB, PDRB, HEGM, DACMM, and TEM modules, we conducted comprehensive ablation studies, both quantitatively and qualitatively. The impact of SEB, PDRB, HEGM, and DACMM was assessed by systematically removing each module individually. Excluding SEB, PDRB, or HEGM resulted in significantly degraded reconstruction performance, as shown in the first three rows of Table V and the first three columns of Fig. 9. Notably, omitting PDRB led to rapidly increasing errors at higher noise levels, underscoring its role in enhancing robustness under noisy conditions. When the DACMM module was removed, the system's performance deteriorated significantly, with MAE and RMSE increasing by 17.65 mm and 18.76 mm, respectively, under low-SNR conditions (fourth row of Table V). Further analysis examined the need to integrate iToF features into DACMM. In the DACMM-a configuration, which corresponds to “ $\times^a$ ” in Table V, we removed the iToF branch, retaining only the RGB input. The results show smaller reductions in MAE and RMSE than when completely removing DACMM, confirming the importance of incorporating iToF features for enhanced performance. We also evaluated the three-stage cascade structure of DACMM by retaining only the final stage (DACMM-b, which “ $\times^b$ ” in Table V). The reconstruction quality degraded largely compared to the full DACMM module (i.e., the last row in Table V), demonstrating the effectiveness of the cascaded structure. Lastly, removing the TEM module

TABLE VI
ABLATION STUDY ON DIFFERENT LOSSES

Datasets	L_{Fisher}	L_{Dis}	L_{Edge}	SNR = 22 dB		SNR = 19 dB		SNR = 16 dB	
				MAE $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	RMSE $\downarrow$
NYU	✗	✗	✗	11.96	21.12	28.38	51.90	68.91	108.58
	✓	✗	✗	10.18	18.66	27.96	43.09	56.64	82.16
	✓	✓	✗	10.77	21.50	29.85	52.03	67.17	102.63
	✓	✓	✓	9.98	18.88	22.41	42.83	45.84	79.78
	✓	✗	✓	12.63	20.53	20.51	30.21	40.50	56.14
	✓	✓	✓	11.77	19.39	20.04	29.39	40.11	55.94
	✓	✓	✓	11.59	18.04	18.42	28.17	29.51	42.23
	✓	✓	✓	9.36	16.50	11.8	18.85	17.53	28.38
4D Light Field	✗	✗	✗	15.53	26.26	35.75	59.22	87.70	123.49
	✓	✗	✗	14.37	25.56	34.76	58.07	68.67	105.98
	✓	✓	✗	14.56	24.05	35.16	59.13	86.89	119.00
	✓	✓	✓	10.97	19.43	27.37	45.23	57.27	88.73
	✓	✗	✓	18.72	29.12	29.61	41.35	55.53	73.50
	✓	✓	✓	17.72	25.43	24.08	38.91	46.28	67.45
	✓	✓	✓	17.84	28.51	23.45	35.88	39.09	54.62
	✓	✓	✓	9.36	16.50	11.8	18.85	17.53	28.38

TABLE VII
COMPARISON OF OUR PROPOSED METHOD WITH A STANDARD TRANSFORMER-BASED ATTENTION MECHANISM

Algorithms	SNR = 22 dB				SNR = 19 dB				SNR = 16 dB			
	MAE	RMSE	Abs.Rel	Sq.Rel	MAE	RMSE	Abs.Rel	Sq.Rel	MAE	RMSE	Abs.Rel	Sq.Rel
Transformer	13.06	19.26	0.0106	0.0965	15.42	21.77	0.0133	0.1324	28.49	45.04	0.0255	0.6719
Ours	9.36	16.50	0.0090	0.0762	11.80	18.85	0.0091	0.1004	17.53	28.38	0.0146	0.2112

and replacing it with three convolutions resulted in noticeable performance degradation, further validating its effectiveness.

2) *Comparison of Different Loss Functions*: In addition to the depth fidelity loss function, which must be used to guide iToF imaging systems in depth reconstruction, we also propose Fisher information guidance loss, multiple measurement discriminative loss, and hybrid edge guidance loss, which can greatly promote the convergence of learnable coding functions and achieve high-quality depth precision with sharper details. To investigate the effect of the components of the loss functions, we remove the Fisher loss, the hybrid edge guidance loss, and the discriminative loss, respectively. As shown in Table VI, among the first four groups of models that exclude the hybrid edge guidance loss, the fourth group achieves the lowest error, demonstrating that the combination of Fisher's information guide loss and multiple measurement differential loss significantly improves reconstruction results. The fifth group, which applies only fidelity loss and hybrid edge guidance loss, shows a deterioration in reconstruction performance, further emphasizing the critical role of Fisher's information guide loss and multiple measurement differential loss. The proposed method, which integrates Fisher information-guided loss, a hybrid edge guidance loss, and a multiple-measurement differential loss, achieves the best reconstruction quality. These results clearly demonstrate the effectiveness of the combined loss functions.

3) To validate the effectiveness of our proposed DACMM module, we conducted an ablation study by replacing it with a strong transformer-based attention baseline, the Multi-DConv Head Transposed Attention (MDTA) from Restormer [70]. As shown in Table VII, our DACMM-based approach significantly outperforms the transformer variant, reducing the MAE by 28.3% and Abs. Rel. by 15.1%. Qualitative comparisons in Fig. 10 further reinforce this conclusion. The error maps reveal that the transformer baseline produces noticeable artifacts along object boundaries and increased texture noise on planar surfaces like walls and floors. In contrast, our method yields smoother

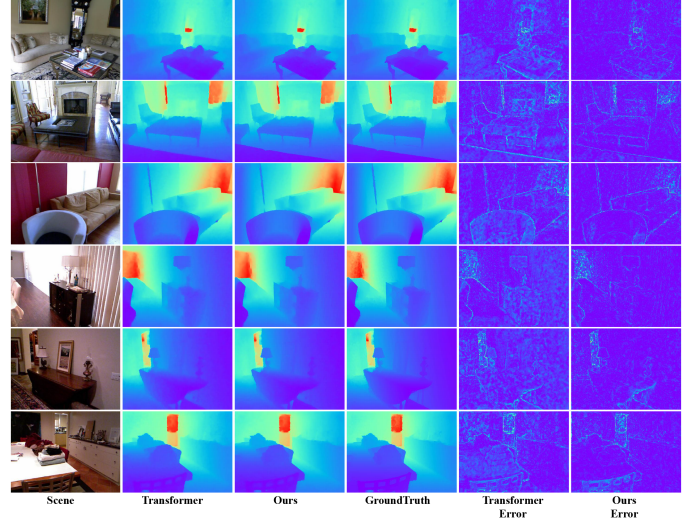


Fig. 10. Visual comparison between the transformer-based attention mechanism and our DACMM module.

TABLE VIII
COMPARISON OF THE EFFECT OF THE FISHER-GUIDANCE TERM UNDER DIFFERENT LOSSES

Datasets	L_{Fisher}	L_{Dis}	L_{Edge}	SNR = 22 dB		SNR = 19 dB		SNR = 16 dB	
				MAE $\downarrow$	fisher	MAE $\downarrow$	fisher	MAE $\downarrow$	fisher
NYU	✗	✗	✗	11.96	509902	28.38	285879	68.91	161923
	✓	✗	✗	10.77	767984	52.03	434397	67.17	248114
	✓	✓	✗	12.63	631699	20.51	357818	40.50	204512
	✓	✓	✓	11.77	473732	20.04	268834	40.11	153831
	✓	✗	✓	10.18	729749	27.96	414671	56.64	237729
	✓	✓	✓	9.98	789442	22.41	449252	45.84	257975
	✓	✓	✓	11.59	698477	18.42	394713	29.51	225189
	✓	✓	✓	9.36	924972	11.8	456436	17.53	261255

reconstructions with significantly fewer errors in these challenging regions. These results suggest that our proposed attention mechanism, which explicitly accounts for the unique spatial characteristics of iToF and RGB data, is more effective for this cross-modality fusion task than a generic, powerful transformer.

4) To quantitatively validate the impact of our proposed Fisher-guidance term (L_{Fisher}), we conducted a comprehensive ablation study as summarized in Table VIII. First, incorporating L_{Fisher} consistently increases the Fisher information across all Signal-to-Noise Ratio (SNR) levels; for instance, at an SNR of 22dB, it boosts the maximum Fisher value by approximately 20%. This information gain directly translates to improved depth estimation, as higher Fisher values are strongly correlated with lower Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). Second, the term significantly enhances robustness. As the SNR drops from 22dB to 16dB, our full model's MAE degrades by only 87%, a stark contrast to the 524% degradation observed in the best-performing model without L_{Fisher} . This highlights its role in ensuring reliable performance in noisy conditions. This performance gain is rooted in the regularizing effect of L_{Fisher} on the learned coding functions. As shown in Fig. 11, our method produces encoding curves with increased saturation that utilize the full available amplitude range, unlike the narrow-band, quasi-periodic waveforms learned without this guidance. Collectively, these findings confirm that the Fisher-guidance term is essential for maximizing measurement

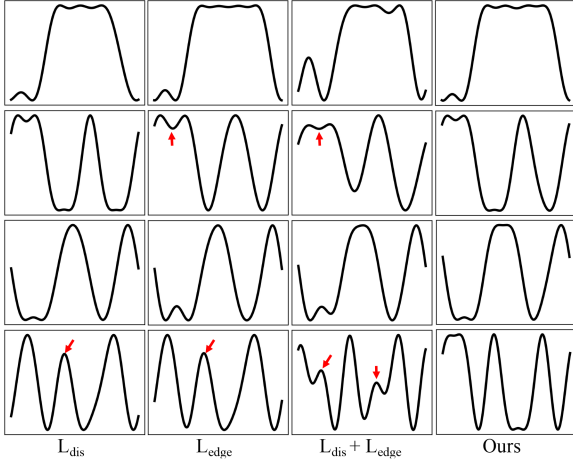


Fig. 11. Visual comparison of coding function under the guidance of different losses.

information, improving depth accuracy, and building a robust imaging system.

F. Evaluation on Real Captured Measurements

1) *Prototype System*: To demonstrate our method with a physical experiment, we built a visual ToF prototype system that can implement imaging with arbitrary coding functions. The learned coding functions are generated by the function signal generator with a maximum bandwidth limit ($f_{\max} = nf_0$) of 250 MHz (DG5252, Rigol). Under this bandwidth constraint, the modulation waveforms are Hamming-windowed to balance theoretical optimality with hardware realizability. We use a 638 nm laser diode with a maximum power of 200 mW (L638P200, Thorlabs) as the light source. With the bias-T coupling circuit (LDM56, Thorlabs), the laser can be modulated by the modulation function. We scan the scene with a galvo-mirror system (GVS012, Thorlabs). The reflected light is focused by a lens (AF NIKKOR, Nikon) onto an avalanche photodiode (APD430 A, Thorlabs). The APD converts reflected light into electrical signals. Then the converted signal is multiplied by the learned demodulation function with a multiplier (AD835, ADI), and the output is further low-pass filtered (EF110, Thorlabs). Finally, the analog signals are sampled, quantized, and translated into digital values by a 16-bit ADC (ADS112C04, TI). Scanning and acquisition of the system are controlled by the Microcontroller (STM32F103, ST) at 2 milliseconds per point. For RGB imaging, an RGB camera (UI-3240CP-C-HQ) is placed coaxially with the scanning galvanometer, through a plane beam splitter, to acquire RGB images of the same scene. The reflected light path is partially transmitted back to the APD through the plane beam splitter, and part of it is reflected to the RGB camera, as shown in Fig. 12.

2) *Plate Depth Accuracy Test*: To evaluate the accuracy of our reconstruction method, we conducted flat-panel reconstruction experiments to obtain depth reconstructions at various depths in real-world scenarios under high ambient light conditions of 9000 lx. A flat plate was placed at depths of 130-160 cm, with measurements taken at 10 cm intervals. These depths were

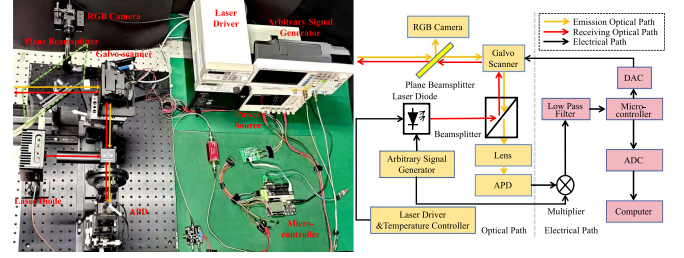


Fig. 12. The hardware implementation of the proposed visual ToF system. Left: Top-down view of the prototype. Right: Corresponding schematic diagram illustrating the signal flow and key components.

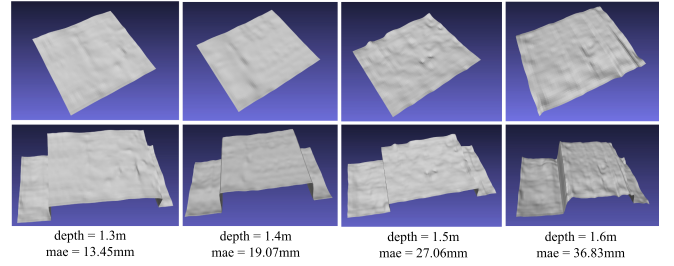


Fig. 13. Reconstruction results of the flat plate at different depths. The first row shows the visualization results of cropping the flat plate areas at different depths, and the second row contains the background depth.

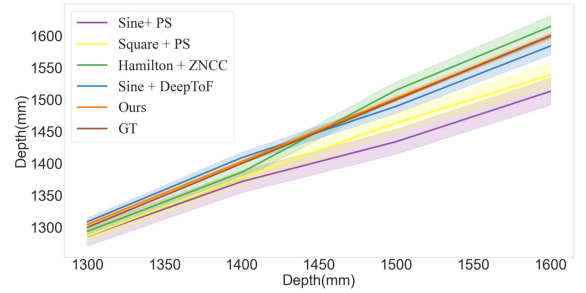


Fig. 14. The mean and variance comparison of different reconstruction methods.

reconstructed using our method. The 3D visualization of the flat plate reconstruction results is presented in Fig. 13. The results indicate that reconstruction performance is better at closer distances, while surface smoothness decreases with increasing depth due to noise interference. For plates at depths of 1.30 m, 1.40 m, 1.50 m, and 1.60 m, the MAEs for different methods are 13.45 mm, 19.07 mm, 27.06 mm, and 36.83 mm, respectively, demonstrating relatively small errors. Fig. 14 provides a comparative analysis with other methods, highlighting the accuracy of our system's depth reconstruction. The figure illustrates the mean and variance of different reconstruction techniques, showing that our method produces depth values most closely aligned with the true values and exhibits the least variance.

3) *Overall Performance Test*: By adjusting the ambient light conditions, we collected iToF measurements under ambient lighting intensities of 9,000 lx (for triangle plaster and foot plaster models) and 10,000 lx (for hand and block models). As shown in Fig. 15, depth imaging was conducted across various scenes. While depth reconstruction results from other methods

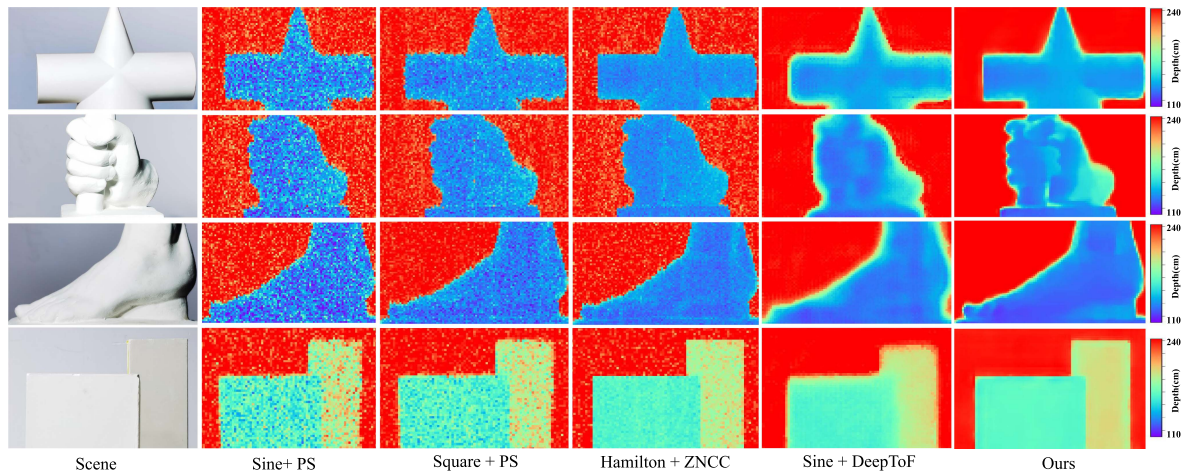


Fig. 15. Performance comparisons on real scene measurements with the state-of-the-art iToF methods, i.e. Sine/Square + PS algorithm [1], Hamilton [7] and Sine + DeepToF [5].

exhibit significant noise, our proposed method effectively suppresses noise and enhances surface smoothness. Furthermore, our method preserves finer details in object features, such as the intricate details at the intersections of the plaster models, the edges of fingers and toes, and the boundaries of plates at varying depths. These results demonstrate the robustness of our method for reliable, high-quality depth reconstruction under high ambient noise.

4) *Resource Efficiency Analysis:* To evaluate our method in resource-constrained settings, we analyzed its power efficiency and depth accuracy against alternative sensing modalities as shown in Fig. 9 in Supplementary material Section V. Our system operates at a low optical power of approximately 0.95 mW, which is comparable to a custom single-beam dToF sensor (1 mW) and lower than a commercial 8×8 dToF sensor (2 mW). In contrast to active sensors, passive RGB-only monocular depth estimation methods like Depth Anything v2 consume negligible illumination power and produce visually detailed depth maps, but they are metrically inaccurate and lack an absolute scale. Our approach resolves this trade-off by fusing metrically informative iToF measurements with visual cues from RGB images. This fusion enables us to achieve high-fidelity, metrically accurate depth reconstruction with fine structural details, while maintaining competitive power consumption, making it well-suited for resource-constrained applications.

VI. CONCLUSION

In this study, we introduce a novel visual ToF imaging paradigm based on a hybrid visual-ToF scheme for precise depth sensing. Building on this paradigm, we design a dual-branch visual ToF depth reconstruction that jointly learns the iToF coding functions and the reconstruction network end-to-end. To further optimize the depth-encoding capability of the learned iToF coding functions, we incorporate a discriminative Fisher-information loss as an optimization guidance. Extensive experiments demonstrate that our method achieves state-of-the-art depth reconstruction performance under high-noise conditions

across the NYU-Depth V2, SUN RGB-D, and 4D Light Field datasets, as well as real-world captured data.

From a broader perspective, our work establishes a general optimization framework that couples visual information with learned ToF coding and reconstruction. While the current study primarily focuses on improving depth reconstruction accuracy and noise robustness, this framework has the potential to support a wider range of ToF imaging scenarios and computational imaging tasks. Future work includes extending the proposed framework to array-based ToF sensors (e.g., ICCD) by optimizing exposure-gating waveforms for more efficient depth imaging over long ranges. Another direction is to address typical ToF challenges, such as multipath interference and occlusions, through joint modeling and learning. In particular, leveraging RGB visual cues for structural and semantic context, together with information-driven optimization of iToF coding functions, may improve the disambiguation of challenging measurements and enhance effective observability under severe multipath interference. Finally, we will explore developing a plug-and-play Fisher-information engine that can be integrated with broader computational imaging modalities, improving robustness and precision for higher-dimensional light-field perception.

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